

TANZINA HOSSAIN

Ph.D. Candidate in Information Science (Data Science) | Finance Educator and Researcher

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EDUCATION AND PROFESSIONAL EXAMINATIONS

Ph.D. in Information Science, Concentration in Data Science

Expected Spring 2027

University of North Texas, USA • GPA: 3.94 / 4.00

- **Dissertation:** *Computational Analysis of Corporate Disclosures: Measuring Change, Identifying Novel Information, and Retrieving Evidence from MD&A.* **Qualifying exam passed April 2026**
- **Selected coursework:** Applied Machine Learning; Natural Language Processing; Discovery and Learning with Big Data; Python and Analytics in Finance; Predictive Analytics

Master of Business Administration (MBA), Finance

2005

University of Dhaka, Bangladesh • GPA: 3.47 / 4.00

Bachelor of Business Administration (BBA), Finance and Banking

2003

University of Dhaka, Bangladesh • GPA: 3.79 / 4.00

CFA Program, CFA Institute

2017–2018

Passed Level I (2017) and Level II (2018)

TEACHING INTERESTS

Investments; Security Analysis and Portfolio Management; Fixed Income Securities; Financial Derivatives; Insurance and Risk Management; Corporate Finance; Financial Markets and Institutions; Data Mining; Data Visualization; Machine Learning.

ACADEMIC APPOINTMENTS AND TEACHING EXPERIENCE

University of North Texas, Denton, TX

Research & Teaching Assistant, Department of Information Science

2023–Present

Courses supported: Research Methods and Analysis; Knowledge Management Tools and Technologies; Information Organization; Information and Knowledge Professions

- Supported instruction and assessment across undergraduate and graduate courses with 200+ combined enrollments.
- Contributed to the development of lecture slides, instructional videos, graphics, handouts, and assessments.
- Delivered lectures on research ethics, data literacy, and applied research methods using business and policy datasets.
- Supported students from varied academic and cultural backgrounds through clear instructions, timely feedback, and individual and group opportunities to participate.

Daffodil International University, Dhaka, Bangladesh

Faculty, Department of Business Administration

2006–2022 (Study Leave)

Associate Professor (2017–2022) • Assistant Professor (2011–2016) • Senior Lecturer (2008–2011) • Lecturer (2006–2008)

Courses taught: Corporate Finance; Financial Management; Financial Markets and Institutions; Financial Derivatives and Markets; Fixed Income Securities; Principles of Finance; Fundamentals of Investments; Security Analysis and Portfolio Management; Insurance and Risk Management; Fundamentals of Banking; Working Capital Management; Public Finance; Real Estate Finance; Financial Analysis and Control; Financial Statement Analysis; Research Methods; Basic Mathematics; Business Statistics.

- Taught undergraduate and graduate finance courses in person, online, and in blended formats and developed digital content for **13** courses in Moodle through DIU's Blended Learning Centre (BLC).
- Designed MBA and BBA curricula and lessons incorporating empirical case studies and data-driven assignments.

- Supervised more than **100** master's theses and undergraduate and graduate internship reports and provided academic, registration, and career advising.
- Used mind mapping, industry visits, Mentimeter, peer instruction, and other interactive learning activities to connect business concepts with practical applications.

Residential Hall Provost, Rawshan Ara Scholar Garden (female student residential hall)**2022**

- Oversaw student welfare, residential discipline, and the academic and administrative coordination of hall residents.

Associate Head, Department of Business Administration**2019–2021**

- Co-managed academic and administrative operations, progress reporting, student affairs, and international partnerships with the department head.

Academic Director, Uttara Campus**2015–2021**

- Directed academic operations across four departments: Business, Computer Science and Engineering, English, and Law, with responsibility for academic quality assurance and faculty performance monitoring.

Dongseo University, Busan, South Korea**Guest Lecturer, Asia Summer Program****2018**

- Taught *Economics of Virtual Currency: A Special Focus on Bitcoin and Blockchain*.

Dhaka City College, Bangladesh**Part-Time Lecturer, Department of Finance and Banking****2005–2006**

- Taught Financial Management, Principles of Accounting, and Banking and Insurance, prepared course materials, assessed student performance, and provided academic guidance.

TECHNICAL AND ANALYTICAL SKILLS

- **Programming and Statistical Software:** Python, R, SQL, Altair AI, Excel, Stata, SPSS, SmartPLS
- **Financial Databases:** WRDS, Compustat, CRSP, SEC EDGAR
- **Data Collection and Management:** Web scraping, APIs, data cleaning, data integration
- **Data Science and Text Analysis:** Machine learning, natural language processing, large language models, retrieval-augmented generation
- **Empirical Methods:** Panel-data analysis, fixed-effects models, event-study methods
- **Visualization and Research Tools:** Tableau, ggplot2, Matplotlib, Git, GitHub, LaTeX

RESEARCH**Peer-Reviewed Journal Articles**

Hossain, T., Malik, M., Hossain, K. S. M. T., & Ryan, S. (2026). Are industry sectors critical for ESG score prediction? Evidence from the U.S. manufacturing and service sectors using machine learning methods. *Journal of Corporate Accounting & Finance*. <https://doi.org/10.1002/jcaf.70052>

Mahmud, I., **Hossain, T.,** Mahmud, A., Akter, R., Rahman, S., & Begum, A. (2025). Predicting fintech adoption: Evidence from an emerging market. *Vision: The Journal of Business Perspective*. <https://doi.org/10.1177/09722629251352696>

Mia, M. A., **Hossain, T.,** Nesa, Z., Saifullah, M. K., Akter, R., & Hossain, M. I. (2025). The impact of board gender diversity on the financing costs of microfinance institutions: Global evidence. *Journal of Financial Reporting and Accounting*, 23(3), 876–901. <https://doi.org/10.1108/JFRA-04-2022-0125>

Hossain, T., & Siddiqua, P. (2024). Exploring the influence of behavioral aspects on stock investment decision-making: A study on Bangladeshi individual investors. *PSU Research Review*, 8(2), 467–483. <https://doi.org/10.1108/PRR-10-2021-0054>

- Muzareba, A. M., **Hossain, T.**, & Nesa, Z. (2023). Insights on investment psychology of individual investors at Dhaka Stock Exchange during the COVID-19 pandemic. *Indonesian Journal of Social Research*, 5(1), 79–90. <https://doi.org/10.30997/ijsr.v5i1.244>
- Shahriar, M. S., Hasan, K. B. M. R., **Hossain, T.**, Beg, T. H., Islam, K. M. A., & Zayed, N. M. (2021). Financial decision making and forecasting techniques on project evaluation: A planning, development and entrepreneurial perspective. *Academy of Entrepreneurship Journal*, 27(4), 1–7.
- Hossain, T.**, & Nesa, Z. (2011). Performance evaluation of mutual funds in Bangladesh. *Journal of Finance and Banking*, 9(1), 157–176.
- Hossain, T.** (2010). Islamic credit card: A prospective in Islamic banking in Bangladesh. *Journal of Banking & Financial Services*, 4(1–2).
- Hossain, T.** (2008). Does categorization influence price behavior of shares? A study on Dhaka Stock Exchange. *DIU Journal of Business and Economics*, 3(1).

Forthcoming Journal Articles and Conference Proceedings

- Rashid, M., Hong, L., **Hossain, T.**, Wang, N., Hossain, K. S. M. T., Ye, G., & Ryan, S. (2026). Fin-Debate: A multi-agent LLM framework for automated fact-checking in the financial domain. *Information Processing & Management (Special issue: Trustworthy Content Governance and Safe Generative AI in the Age of Large Language Models)*.
- Rashid, M., **Hossain, T.**, Song, X., & Hong, L. (2026). Task-aware methods for multilingual financial NLP. In *Proceedings of the 11th Workshop on Financial Technology and Natural Language Processing (FinNLP 2026)*. Association for Computational Linguistics.

Book

- Hossain, T.** (2012). *Remittance issues & practices: A Bangladesh perspective*. LAP LAMBERT Academic Publishing. ISBN 978-3-659-10681-1.

Selected Presentations

- Hossain, T.** (2026, July 24). Beyond boilerplate: Contextual embeddings for measuring economic change in MD&A disclosures. *College of Information Summer Research Day*, University of North Texas.
- Hossain, T.**, Malik, M., & Ryan, S. (2025, February). Predicting ESG ratings for U.S. industries using machine learning: A data-driven approach to Environmental, Social, and Governance evaluation. *Multi-disciplinary Information Research Symposium*, University of North Texas.
- Mia, M. A., **Hossain, T.**, Nesa, Z., Saifullah, M. K., Akter, R., & Hossain, M. I. (2022, March). How board gender diversity affects financing costs of microfinance institutions: Global evidence. *International Conference on Business Management, Macroeconomic Applications and Entrepreneurship Principles & Practices for Sustainable Development*, Daffodil International University.
- Hossain, T.**, & Siddiqua, P. (2021, June). Effect of behavioral characteristics on stock investment decision-making: A study on individual investors in Dhaka Stock Exchange. *International Symposium on Current Approaches & New Trends in Social Sciences*, Toros University, Turkey.

Working Papers and Research in Progress

- Hossain, T.**, Hossain, E., Hossain, K. S. M. T., & Ryan, S. When words change meaning: Contextual embeddings for measuring MD&A disclosure change.
- Hossain, T.**, & Hossain, E. Does artificial intelligence reduce corporate litigation risk? Evidence from firm-level litigation.
- Hossain, T.**, Malik, M., & Ryan, S. Peer-adjusted novelty and the informational value of MD&A revisions.
- Hossain, T.**, & Ryan, S. Finding What Changed: Revision-Aware Retrieval of Year-over-Year MD&A Changes.

TEACHING, MENTORING, AND COMMUNITY ENGAGEMENT

University of North Texas

Student Coordinator, Data Science & AI Talk Series	2026
Counselor, NexGen Data Leadership Summer Camp	2026
Instructor, Storytelling with Data Summer Camp	2025-2026

Daffodil International University

Convener, Mock Stock Trading	2017-2019
Instructor, Advanced Trading Affairs Learning of Capital Market (workshop)	2011-2013
Instructor, Capital Market ABC and Personal Investment (workshop)	2010-2013
Convener, Finance Exposure Project Fair	2016-2018
Convener, Business Fest@UC	2016
Invited Guest, BLC: NOW & NEXT	July 19, 2020
Instructor, Executive Presentation Skill Development (workshop)	2014-2015

LEADERSHIP AND SERVICE

Ad Hoc Reviewer

2021-Present

Acta Psychologica, Elsevier (2025); Journal of Library Resource Sharing, Taylor & Francis (2025); Journal of Financial Reporting and Accounting, Emerald (2024); Journal of Information and Knowledge Management, World Scientific (2023); International Journal of Research and Innovation in Social Science (2021–2024).

University of North Texas

Founder and President, Women in Data Science (WinDS)	2025–Present
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Daffodil International University

Co-Chair, International Conference on Business Management	2022
Co-Convener and Host, Virtual Postgraduate Research Summit	2022
Member, Complaint Committee to Prevent Sexual Harassment	2016–2022
Member, Academic Council	2015-2021
Member, Curriculum Development Committee	2010–2021
Convener, Examination Committee	2015

HONORS AND AWARDS

Lis and Philip Turner Outstanding Student Paper/Project Award

2025

Department of Information Science, University of North Texas

- *Awarded for Predicting ESG ratings for U.S. industries using machine learning: A data-driven approach to Environmental, Social, and Governance evaluation.*